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CHE 321; Spring - 2020; Quiz - 2 100 Points

1. (15 points) A first order reaction is carried out in 2 equal sized reactors. Select all options that are true about the overall conversion of the reaction:
- a. higher with 2 PFRs in series than with 2 PFRs in parallel.
 - 5 ☒ b. Changes with the order of PFR and CSTR (PFR-CSTR or CSTR-PFR).
 - c. Remain the same for 2 PFRs in series or 2 PFRs in parallel.
 - ☒ d. lower with two CSTRs in series than with two CSTRs in parallel.
2. (15 points) The most **unsuitable** reactor for carrying out reactions in which high reactant concentration favors high yields is
- a. series of CSTR
 - 15 ☒ b. one CSTR
 - c. plug flow reactor
 - d. series of PFRs
3. (15 points) 'N' plug flow reactors in series with a total volume 'V' gives the same conversion as a single plug flow reactor of volume 'V' for _____ order reactions.
- a. third
 - b. first
 - 15 ☒ c. any
 - d. second
4. (15 points) The ratio of volume of CSTR to the volume of P.F.R. (for identical flow rate, feed composition and conversion) for zero order reaction is
- ☒ a. 1
 - 15 b. ∞
 - c. 0
 - d. >1

5. (15 points) Explain the answer above

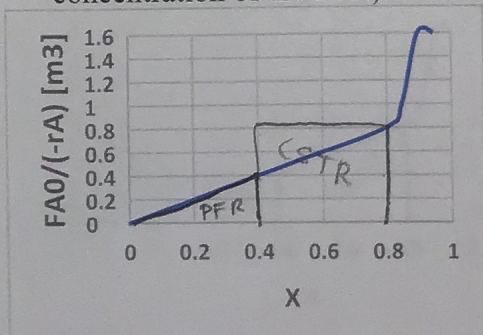
15 At zero order, the concentrations would be the same, so the volume would be equal.

$$\frac{\text{CSTR}}{\text{PFR}} = 1$$

$$V = \frac{F_0 - F}{r_A} = \frac{F_0 - F}{kC_A}$$

$$\text{PFR} = \frac{F_0 - F}{r_A}$$

6. (25 points) The following Levenspiel plot shows the response of a chemical reaction. A PFR and then a CSTR in series show a conversion of 0.4 and 0.8 (based on the entry concentration of the PFR). Find the volumes of the PFR and the CSTR.



$$\frac{1}{2}(0.4)(0.4) = 0.08 \text{ m}^3 \rightarrow \text{PFR}$$

$$\frac{1}{2}(0.4)(0.4) = 0.08 + 0.08 + 0.16 = 0.32 \text{ m}^3 \rightarrow \text{CSTR}$$

$$0.8 \times 0.4 =$$